

ELECTRIC SECTOR CONTROL-COMPANY FUNCTIONAL CLASS

Functional Capability Design Basis: Physical, Human, and Software-Data Layers

DBA Consequence Envelope and Military Action Extension — Full Three-Layer Instrument

DBA-ES-CntlCo: Balancing Authority, Reliability Coordinator, and Transmission Operator

A Functional Class Instrument Under

DBA-ES: Electric Sector Design Basis Framework (v0.8)

DBA-EN-SF: Energy Sector Framework (v0.8)

Predecessor: DBA-ES-CntlCo v0.1 (DB Layer)

Version 0.4 — DRAFT

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For review by ISO/RTO operations staff, NERC reliability analysts, and balancing authority engineers,
FERC staff and policy officials responsible for bulk electric system reliability governance.

Section 1 — Purpose, Instrument Type, and Version Status

1.1 Purpose and Instrument Status

This document completes the three-layer DBA-ES-CntlCo instrument by adding the DBA consequence envelope and the DBA-MA military action extension to the DB layer established in v0.1. It addresses Gap 2.1 in the DBA-EN series gap register and satisfies the sequencing prerequisite for DBA-ES-DI-FC5 v0.3, which requires a complete CntlCo consequence envelope before its own DBA and MA extension can be drafted.

The analytical architecture confirmed in the Discussion prior to drafting is: a unified functional class with a single consequence envelope; Situational Awareness Blackout Duration (SABD) as the governing consequence metric; and SCADA/EMS integrity attack as the governing DBA-MA scenario. These three architectural choices are developed in full across Sections 6 through 9.

1.2 Governing Metric — Situational Awareness Blackout Duration (SABD)

SABD is defined as the continuous interval between loss of state estimator validity and restoration of coordinated real-time operational authority. It is the CntlCo analog to RT_HVLLC in the facility-class instruments, where RT_HVLLC measures the time to restore a facility's physical capability following destruction of its most difficult-to-replace component, SABD measures the time to restore the CntlCo's functional capability following loss of its governing operational tool — the state estimator.

SABD begins at the moment the state estimator is confirmed non-convergent, produces outputs flagged as unreliable, or is taken out of service due to suspected data integrity compromise. SABD ends at the moment a qualified system operator confirms the state estimator is producing convergent, validated results and resumes issuing dispatch instructions based on its output. The interval includes: the time to identify the state estimator failure mode, the time to implement backup operating procedures, the time to restore data feed integrity, and the time to validate state estimator convergence after restoration.

SABD is a functional metric, not a physical one. It cannot be derived from equipment lead times alone. It depends on operator training currency, backup procedure availability, data feed restoration capability, and the mode of state estimator failure. A state estimator failure from random hardware faults has a different expected SABD than a state estimator failure from systematic adversarial data injection — the latter requires validation of the data environment before the state estimator can be safely restarted.

1.3 Position in Series and Sequencing

Document Number: DBA-ES-CntlCo-001

Version: 0.2 — Full three-layer instrument

Parent Document: DBA-ES: Electric Sector Design Basis Framework (v0.8)

Series Grandparent: DBA-EN-SF: Energy Sector Framework (v0.8)

Predecessor: DBA-ES-CntlCo v0.1 (DB layer only)

Sequencing: Completion of this instrument satisfies the prerequisite for DBA-ES-DI-FC5 v0.3 (DI-1/DI-2 in that instrument's open item register). FC5 v0.3 may reference this instrument's consequence envelope, SABD metric, and MA extension without separately establishing them.

Section 2 — Three-Layer Design Basis Structure (Retained from v0.1)

The three-layer design basis structure — Physical Layer (PL), Human Layer (HL), and Software-Data Layer (SL) — is fully established in DBA-ES-CntlCo v0.1 and is not re-derived here. The postulated event tables for all three layers (PL-1 through PL-5, HL-1 through HL-5, SL-1 through SL-5) are carried forward without substantive change. The cross-sector cascade category (XC) is carried forward from the DBA-ES sector parent.

This section provides a summary reference. The governing text remains in v0.1 §§ 3 through 7. Where v0.1 and v0.2 conflict, v0.2 governs.

2.1 Layer Summary

Layer	Governing Failure Mode	v0.1 Event Codes	DBA Extension (this document)
Physical (PL)	Control center / BCC unavailability; communications infrastructure failure	PL-1 through PL-5	IC-2 (coordination failure) governed by PL-4 as the initiating condition
Human (HL)	Operator staffing below minimum; cognitive overload; training currency lapse; access denial; authority chain loss	HL-1 through HL-5	IC-2 (coordination failure) includes HL-2 cognitive overload as compound initiating condition
Software-Data (SL)	State estimator non-convergence; SCADA feed loss; AGC data corruption; cyber intrusion; systematic bad data injection	SL-1 through SL-5	IC-1 (SABD) is the governing SL consequence. MA-1 (state estimator corruption) is the governing adversarial SL event
Cross-Sector	FC instrument data feed loss	XC-1 through	IC-3 (compound) is the

(XC)	producing SL-layer degradation; CntlCo functional loss producing FC-level consequences.	XC-4	governing XC consequence event
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Section 3 — DBA Consequence Envelope

3.1 Unified Functional Consequence Envelope

The CntlCo consequence envelope is unified across the BA, RC, and TOP functional triad. This is architecturally appropriate because the functional triad operates as a single-layered authority structure in real time: TOP implements what BA dispatches, and RC overrides both when system reliability requires. A consequence analysis that disaggregates the triad into separate consequence envelopes would produce artificially narrow consequence estimates for each function while missing the compound consequence of simultaneous degradation across the authority chain.

The unified envelope is bounded by three initiating conditions that span the single-function and compound failure space. IC-1 addresses the state estimator — the governing analytical tool of the SL layer. IC-2 addresses inter-entity coordination — the governing function of the RC authority chain. IC-3 addresses the compound failure — the Uri paradigm in which both situational awareness and coordination degrade simultaneously.

3.2 Initiating Conditions

Code	Initiating Condition	Governing Layer(s)	Consequence Tier
IC-1	Complete loss of EMS/SCADA visibility — state estimator non-convergent or taken out of service due to suspected data integrity compromise. BA is operating on backup procedures without a validated system state picture.	SL (primary); HL (secondary — operator workload elevation)	Tier 2 (grid-level): BA can issue manual dispatch but cannot verify system state. Tier 3 if a disturbance develops during the SABD interval.
IC-2	Loss of inter-entity coordination — RC-to-BA authority chain communications failure or RC functional unavailability. BA areas cannot coordinate reserve sharing, curtailment, or emergency directives across area boundaries.	PL (primary — PL-4); HL (compound — HL-5); SL (secondary)	Tier 2 (grid-level): Local situational awareness intact, but coordinated response to inter-area events is unavailable.

			Compound with IC-1 produces Tier 3.
IC-3	Compound initiating condition — IC-1 and IC-2 simultaneously. BA is operating without a validated state picture and without RC coordination. Governing compound scenario: developing disturbance during the period of combined SABD and coordination failure—Uri paradigm empirical anchor.	All three layers simultaneously	Tier 3 (cross-sector cascading): governing CntlCo consequence. Gas-electric cascade acceleration if IC-3 occurs during the high gas-generation reliance period.

3.3 Consequence Taxonomy

The three-tier consequence taxonomy established in DBA-EN-SF §2.3 applies to the CntlCo consequence envelope. Tier designations for the CntlCo functional class are defined as follows.

Tier	CntlCo Consequence Definition	Governing IC	Acceptance Criterion Category
Tier 1	CntlCo functional degradation confined to the BA area — BA unable to maintain automatic generation control or issue validated dispatch instructions. Still, RC and adjacent BA areas are unaffected and able to provide coordination support.	IC-1 in isolation	SABD within design basis maximum; manual operations procedures activated; RC notified within notification time criterion.
Tier 2	Bulk electric system reliability impact across multiple BA areas — RC coordination failure removes emergency response authority above the BA level; N-1 contingency management capability is degraded or absent across the RC area.	IC-2; IC-1 with developing disturbance	Emergency procedures activated; NERC Event Analysis notification criterion triggered; emergency load shedding authority confirmed at the BA level.
Tier 3	Cross-sector cascading — CntlCo functional loss during a developing disturbance produces or accelerates a cascade affecting critical services, gas-fired generation curtailment, or load shedding affecting life-safety services.	IC-3 compound	All Tier 2 criteria plus: gas-electric cascade protocol invoked; Federal Coordination Threshold notification issued; cross-sector consequence elevation rule applied per DBA-EN-SF §3.4.

3.4 SABD Acceptance Criterion Framework

SABD acceptance criteria cannot be established at the series level because the acceptable SABD duration depends on the BA area's specific topology, the contingency risk profile at the time of state estimator loss, and the backup operating procedures available. The series-level framework establishes the criterion structure; site-specific calibration is required at implementation.

SABD Zone	Condition	Required Response
SABD Zone 1 (0 to T ₁)	State estimator non-convergent, but data feeds intact. Backup operating procedures activate. The operator is working from the last valid state picture.	Manual operations mode activated. RC notified. State estimator recovery initiated. If T ₁ exceeded, escalate to Zone 2.
SABD Zone 2 (T ₁ to T ₂)	State estimator recovery not achieved within T ₁ . Data feed integrity unconfirmed. Operator situational awareness degraded beyond manual procedure coverage.	Emergency operating procedures activated. Emergency load shedding authority confirmed—neighboring BA areas notified. RC assumes coordinating authority. If T ₂ exceeded, escalate to Zone 3.
SABD Zone 3 (beyond T ₂)	State estimator unavailable beyond T ₂ —CntlCo functional degradation at risk of producing Tier 3 consequence.	Federal Coordination Threshold notification. Emergency interchange adjustments with adjacent BA areas. Cross-sector consequence elevation rule evaluated per DBA-EN-SF §3.4. Mutual aid CntlCo support invoked.

T₁ and T₂ calibration values are established in the site-specific design basis implementation. Series-level guidance: T₁ is bounded by the NERC EOP-011-2 notification requirement timeline. T₂ is bounded by the estimated cascade initiation timeline for the BA area under the governing contingency set. Series-level quantitative bounds are established in DBA-ES-MA-QB v0.1 §2: T₁ ≤ 30 minutes (series upper bound); T₂ ≤ 4 hours (series upper bound). BA-class-differentiated recommended values are provided in DBA-ES-MA-QB v0.1 Table 2.1. Site-specific calibration must produce values at or below the series bounds; implementations should target materially tighter values where BA topology, gas generation penetration, or peak-to-reserve-margin ratios warrant. DI-7 is closed by DBA-ES-MA-QB v0.1 §2.

Section 4 — RC and TOP Consequence Extensions

4.1 RC Consequence Extension

The RC consequence extension applies the unified envelope to the RC oversight function. RC functional loss transforms IC-2 from a coordination degradation into a coordination elimination: the authority chain above the BA level is removed. The RC's consequence is not facility-level

(the RC does not directly operate transmission infrastructure) but authority-chain-level — its loss removes the capacity for coordinated emergency response across BA boundaries.

The governing RC consequence is the RC area-wide degradation of N-1 contingency management capability: with no RC authority above the BA level, individual BAs must make uncoordinated responses to inter-area disturbances. The 2003 Northeast Blackout is the governing empirical anchor — the absence of effective RC-level coordination was a primary enabling factor in the cascade propagation.

RC authority chain communications design basis: The mechanism for maintaining RC-to-BA authority chain communications during PL-4 conditions must include a minimum of one backup communications path that is physically and logically independent of the primary SCADA/EMS communications infrastructure. This resolves DI-3 from v0.1. The specific backup path design is established in the site-specific implementation.

4.2 TOP Consequence Extension

The TOP consequence extension applies the unified envelope to the local switching authority function. TOP functional loss removes the authority to execute transmission switching operations within the TOP area — the physical control actions that implement BA dispatch instructions and RC emergency directives at the transmission level. IC-1 at the TOP level means the TOP is executing switching operations without a validated system state picture; IC-2 at the TOP level means the TOP is receiving no emergency directives from the BA or RC authority chain.

The governing TOP consequence is the inability to isolate a faulted transmission element within the design basis switching time during a period of simultaneous CntlCo functional degradation. The compound scenario — a transmission fault requiring rapid isolation occurring during SABD Zone 2 or Zone 3 — is a Tier 2 consequence at minimum and Tier 3 if the fault propagates to multiple elements before isolation.

Section 5 — Recovery Architecture

5.1 Recovery Architecture Principles

CntlCo recovery architecture differs from facility-class recovery architecture in one fundamental respect: there is no equivalent of the RT_HVLLC lead-time constraint. The CntlCo function does not depend on the delivery of long-lead physical components — it depends on the restoration of data integrity, operator availability, and communications. Recovery is therefore procedural and functional rather than logistical.

The recovery architecture has three phases corresponding to the SABD zones: Zone 1 recovery (state estimator restoration from backup data); Zone 2 recovery (manual operations mode transition and RC mutual aid); Zone 3 recovery (full functional reconstitution with validated data environment).

5.2 Phase Recovery Requirements

Recovery Phase	SABD Zone	Primary Recovery Action	Success Criterion
Phase 1 — Restore	Zone 1	State estimator recovery: restart from the last convergent solution, restore data feed integrity, and revalidate the topology model. If the state estimator cannot be recovered, transition to Phase 2.	State estimator convergent with less than [site-specific] percent bad data rejected. Operator confirms system state picture consistent with last known good state.
Phase 2 — Degrade-Maintain	Zone 2	Manual operations mode activation: operators issue dispatch instructions based on the last valid state picture and real-time voice communications with field operators. RC mutual aid invoked. Emergency load shedding authority confirmed.	Stable manual operations confirmed. RC mutual aid established. No uncontrolled frequency or voltage excursion during the manual operations interval.
Phase 3 — Reconstitute	Zone 3	Full functional reconstitution: data environment validation completed, state estimator restarted with clean data feeds, operator workload normalized, backup operating procedures stood down.	State estimator converges on validated data—operator-issued dispatch instructions replaced by AGC. RC coordination restored. SABD clock stopped.

5.3 BCC Transition as Recovery Path

Backup control center (BCC) transition is the governing recovery path for PL-1 and PL-2 initiating conditions. The BCC transition is not a recovery from SABD — it is a recovery of the physical platform on which the CntlCo function runs. A successful BCC transition does not automatically terminate SABD. If the state estimator at the BCC cannot synchronize to the current system state within the BCC transition time criterion, the SABD clock continues to run at the BCC.

This distinction matters analytically: a CntlCo that successfully transitions to its BCC within the design basis transition time but cannot restore state estimator convergence is still in SABD Zone 1 or Zone 2 regardless of the BCC transition success. The BCC transition time criterion and the SABD T₁ criterion are separate acceptance criteria that must both be satisfied for the CntlCo to exit the consequence envelope.

5.4 MA Communications Design Basis

DBA-ES-MA-QB v0.1 §5 establishes the MA communications design basis as a sector-level requirement applicable to all DBA-ES FC instruments. The following instantiates those requirements for the CntlCo functional class. The MA communications baseline condition — primary SCADA and voice communications channels are unavailable — is the assumed design basis condition for all MA scenario analysis in this instrument. This is consistent with the coherence gap identified in DBA-ES-MA-SF v0.2 §7.1 and resolved at the sector level by DBA-ES-MA-QB v0.1 §5 (OI-1 closure).

CB-1 (Backup channel): The CntlCo must maintain at least one backup communications channel to each field facility in its area that is architecturally independent of the primary SCADA/EMS communications infrastructure. Acceptable backup channels: dedicated licensed radio, satellite communications, or physically diverse fiber. The CB-1 requirement at the field facility level is also a CntlCo design basis requirement: CB-1 compliance at field facilities is the prerequisite for backup data feed availability to the state estimator during SABD Zone 1 recovery. A CntlCo whose field facilities do not satisfy CB-1 cannot assume backup data feeds will be available for state estimator recovery within T_1 .

CB-2 (Local authority): Field operators must have pre-authorized authority to execute facility-level emergency response procedures without CntlCo direction. The design basis must not require CntlCo authorization for any field-level emergency action whose delay — under conditions where the CntlCo is unreachable — would produce a consequence tier escalation.

CB-3 (Command Broker continuity): The MA communications degradation condition does not suspend the Command Broker obligation for above-Bright-Line Gen AI advisory tools. The licensed system operator on duty remains the mandatory Command Broker regardless of communications status. Above-Bright-Line Gen AI output must not be actioned without licensed operator confirmation, even when primary communications are degraded. Pending the CB-UC instrument (DI-5), no workaround to this requirement is permitted. Communications degradation is a denied condition (FD-BL-D1 §7). The governing foundational instruments — FD-BL (placement), FD-BL-D1 (mode of discharge), FD-BL-D2 (indication integrity), and FD-EV (envelope) — are applied in this annex by reference.

CB-4 (Verification cadence): All backup communications channels must be tested at a defined interval specified in the CntlCo's design basis implementation. Channels installed but untested are not a design basis asset. The test condition must simulate the MA baseline: primary channel unavailable, backup channel as sole path.

CB-5 (Degraded-mode procedures): The CntlCo's emergency operating procedures must include a degraded-communications mode executable without field facility contact. Procedures must address: (a) autonomous CntlCo actions when primary communications to field facilities are lost; (b) backup channel activation sequence; (c) minimum operational state maintained pending communications restoration; and (d) the threshold at which the CntlCo initiates controlled load shedding rather than continue operating in degraded-communications mode without a validated system state picture.

Section 6 — DBA-MA Military Action Extension

6.1 MA Extension Applicability

The military action extension applies the four MA characteristics — intentionality, persistence, precision, and denial — to the CntlCo consequence envelope established in Section 3. The application is not uniform across the four characteristics: precision and denial are dominant for the CntlCo because the governing adversarial scenarios do not require physical destruction of infrastructure. A sophisticated adversary targeting the CntlCo function can achieve Tier 3 consequences without destroying a single physical asset.

6.1.1 MA Applicability Assessment (*Four-Filter Framework*)

Per DBA-ES-MA-SF v0.2 §1.4 and DBA-ES-MA-QB v0.1 §5, each DBA-ES FC instrument must present a structured four-filter applicability assessment at the point where the operator first engages with the MA extension. The assessment must be documented; a conclusion of non-applicability requires documented reasoning against each filter and is a reviewable professional judgment, not a unilateral exemption. For the CntlCo functional class, the assessment is resolved at the class level — all CntlCo facilities satisfy the MA design basis applicability criteria by definition of their function.

Filter 1 — Geolocation relative to defense-critical loads: **SATISFIED BY DEFINITION**. A Balancing Authority, Reliability Coordinator, or Transmission Operator serves all loads in its area, which by definition includes defense-critical loads, military installations, and continuity-of-government facilities in its service footprint. The CntlCo's functional loss degrades service to all loads in the affected area without distinction.

Filter 2 — Position in strategic transmission corridors: **SATISFIED BY DEFINITION**. The CntlCo function is not a node in a transmission corridor — it is the control authority for all transmission corridors in its area. Its loss removes operational authority over every strategic corridor in the affected footprint simultaneously.

Filter 3 — Substitutability: **LOW**. CntlCo mutual aid — the invocation of RC coordinating authority and neighboring BA operational support — is the substitutability mechanism. Its availability under MA conditions is explicitly addressed in §6.5 and §5.3 of this instrument. Mutual aid is available in Zone 2 and Zone 3 but does not fully substitute for the local CntlCo function during the SABD interval; the substitute's effectiveness is reduced by the same communications degradation conditions that affect the primary CntlCo.

Filter 4 — Adversary targeting logic: **HIGH PRIORITY TARGET**. The CntlCo is the sector's operational nervous system. An adversary targeting the electric sector for maximum cascade effect will identify CntlCo functional degradation as a force multiplier: physical attacks on FC2, FC3, and GC-FC4 nodes are more consequential when the CntlCo cannot see, coordinate, or respond to them. DBA-ES-MA-SF v0.2 §4 identifies the CntlCo as a member of every high-consequence pattern attack combination.

Conclusion: The MA design basis applies to all CntlCo facilities. No facility-level exemption assessment is required. The MA extension sections of this instrument apply without a threshold qualification.

This is the defining characteristic that distinguishes the CntlCo MA extension from all other DBA-ES MA extensions. Every other instrument's governing MA scenario involves physical destruction of a long-lead component (FC2 LPT, FC3 converter transformer, FC6 supercritical rotor). The CntlCo governing the MA scenario involves the corruption of information — the state estimator's picture of the system — without any physical destruction. The consequence duration is bounded by SABD, not by RT_HVLLC.

6.2 Four MA Characteristics Applied to CntlCo

MA Characteristic	Application to CntlCo	Governing Concern
Intentionality	The adversary targets the state estimator specifically, not the control center generally. The attack is designed to produce a specific false system state — one that causes the BA to take a damaging action — rather than to deny service.	Intentionality distinguishes MA-1 from random equipment failure: a random state estimator failure produces a known-bad picture that operators recognize as unreliable. An intentional attack produces a plausible-but-false picture that operators may act on. The adversary's goal is to cause the operator to take the wrong action, not to make the operator unable to act.
Persistence	The adversary maintains the false data injection over a sufficient period to cause the operator to commit to a damaging dispatch decision before the false picture is identified. Persistence also applies to MA-2 (SCADA availability attack): denial of SCADA availability over a period that exceeds the manual operations mode capability window.	For MA-1, persistence is measured in minutes to hours — the time required to cause a dispatch decision that cannot be reversed before cascade onset. For MA-2, persistence is measured against the SABD T ₂ criterion: an adversary who can maintain SABD Zone 3 conditions past T ₂ achieves Tier 3 consequence without physical destruction.
Precision	The adversary injects false data at specific measurement points that produce a false system state consistent with normal operations, high enough data quality that bad data detection algorithms do not flag the manipulation. The false picture is precisely targeted to cause a specific	Precision is the MA characteristic that makes MA-1 more dangerous than MA-2. An adversary with sufficient precision can cause the BA to take a harmful action while the operator believes the system is

	incorrect dispatch action.	operating normally. The operator's confidence in the state picture — which is the product of the EMS's bad data detection algorithms — becomes the attack surface.
Denial	MA-1 (state estimator corruption) achieves denial by making the state estimator's output unreliable without making it obviously wrong. MA-2 (SCADA availability) achieves denial directly by removing data feed availability. Both modes deny the operator access to a trustworthy system state picture.	Denial in the CntlCo context is denial of situational awareness, not denial of physical capability. The CntlCo's physical infrastructure may be fully intact during an MA-1 or MA-2 attack. The operator has all the tools needed to manage the system, but the picture those tools are producing is false or absent.

6.3 Governing DBA-MA Scenario — MA-1: State Estimator Corruption

MA-1 is the governing DBA-MA scenario for the CntlCo. The adversary targets the SCADA data environment to inject false measurements at specific points in the system that cause the state estimator to produce a plausible but incorrect system state picture. The false picture is designed to cause the BA to take a specific harmful dispatch action — over-curtailment of reserves in one area, under-curtailment in another, or incorrect redispatch that positions the system near a stability boundary before the false picture is identified.

The attack sequence under MA-1 proceeds through three phases. In Phase 1 (Reconnaissance), the adversary maps the measurement topology of the state estimator — identifying which measurement points, if corrupted simultaneously, will cause the state estimator to converge on the target false state without triggering bad data detection. This requires knowledge of the redundancy structure of the measurement network and the bad data detection thresholds in use. In Phase 2 (Injection), the adversary corrupts the identified measurement points simultaneously, causing the state estimator to converge on the false picture. The injection is designed to be persistent — the corrupted measurements must remain consistent with each other to avoid triggering bad data detection algorithms during the SABD window. In Phase 3 (Exploitation), the adversary waits for the BA to take the target dispatch action based on the false picture. The exploit is complete when the BA has committed to a dispatch configuration that cannot be reversed before cascade onset.

The governing MA-1 consequence is a Tier 3 cross-sector cascading event: the BA, acting on the false state picture, dispatches the system into a configuration that produces an N-1 contingency it cannot recover from within the cascade initiation timeline. The false picture prevents the BA from recognizing the impending contingency until the cascade is already developing.

6.4 Secondary DBA-MA Scenario — MA-2: SCADA Availability Attack

MA-2 is the secondary bounding DBA-MA scenario. The adversary denies SCADA data availability to the BA — either by attacking the communications infrastructure serving data feeds (PL-4 as MA target) or by attacking the EMS/SCADA software directly (SL-2 as MA target). Unlike MA-1, the operator knows the system picture is unavailable. The BA transitions to manual operations mode; the SABD clock begins at IC-1.

MA-2 is less dangerous than MA-1 operationally because the operator knows the picture is gone — manual procedures activate, RC is notified, and the system operates in a known-degraded mode. The adversary's goal under MA-2 is persistence: to maintain SABD Zone 3 conditions long enough that a naturally occurring contingency (or a compound physical attack) finds the BA without a validated system state. The MA-2 adversary uses the CntlCo's own SABD vulnerability as the exploit window.

The governing MA-2 consequence is a Tier 2 or Tier 3 event, depending on whether a significant contingency occurs during the sustained SABD interval. MA-2 in isolation, without a compounding physical event, is a Tier 2 consequence. MA-2, combined with a targeted generation or transmission outage during the SABD interval, is a Tier 3 compound MA scenario.

6.5 RC and TOP MA Extensions

The RC MA extension applies MA-1 and MA-2 to the RC authority chain. The governing RC MA scenario is a coordinated attack that simultaneously degrades multiple BA state estimators within an RC area — causing each BA to operate on a different false picture. In contrast, the RC coordination channel is simultaneously degraded (IC-3 compound under MA conditions). This is structurally analogous to the IC-3 DBA initiating condition but with the four MA characteristics applied: the multiple simultaneous BA state estimator corruptions are intentional, persistent, precise, and designed to deny RC-level coordination.

The TOP MA extension applies precision and denial to the TOP's switching authority. The governing TOP MA scenario is a false data injection that causes a TOP to execute a switching operation that creates a fault rather than isolating one — the adversary corrupts the measurement picture at the TOP level to cause a harmful switching action. This is a facility-specific MA extension; the DBA-ES-TX-FC2 instrument carries the primary analysis for EHV substation switching operations under MA conditions.

Section 7 — Dual Design Basis Architecture and AI Governance

7.1 AI Governance Interaction Points

The CntlCo carries three AI governance interaction points that do not appear in any other DBA-ES instrument. These points are established here as series firsts and may be propagated to other instruments as appropriate.

7.1.1 State Estimator ML Augmentation

Modern EMS state estimators incorporate ML-augmented bad data detection, topology processing, and observability analysis. These ML components are deterministic in the sense that they produce consistent outputs from consistent inputs — they are formally testable, and their behavior can be validated against known system states. Where ML output is limited to advisory flagging and state estimation inputs, these components operate below the Bright Line; no Command Broker is required for ML advisory output in that configuration.

The Bright Line condition for state estimator ML augmentation is that ML output must be limited to advisory flagging and state estimation inputs — the ML system cannot trigger automatic redispatch or automatic load shedding without a human operator confirmation step. If an ML-augmented state estimator is deployed in a configuration where its output automatically triggers AGC setpoint changes without operator confirmation, the ML system has crossed the Bright Line and requires Command Broker governance.

7.1.2 Gen AI Operator Decision Support

Gen AI tools deployed in the CntlCo environment for operational decision support — situational awareness synthesis, emergency procedure guidance, predictive dispatch recommendations — are non-deterministic and operate above the Bright Line. Command Broker governance is mandatory for all Gen AI tools used in real-time operations.

The Command Broker for CntlCo Gen AI tools is the licensed system operator on duty. This is the natural Command Broker for two reasons. First, NERC system operator certification establishes the qualification standard: the operator has demonstrated the domain knowledge required to evaluate Gen AI output against known system behavior. Second, the operator's legal accountability under NERC's rules of procedure provides the governance framework: the operator is personally accountable for dispatch decisions, which creates the accountability structure required for above-Bright-Line AI tool use.

A critical constraint follows from this: a Gen AI tool that advises a licensed system operator on emergency procedures during a developing cascade is above the Bright Line. The Command Broker cannot be a non-licensed person — a supervisor, manager, or IT administrator. The licensed operator on duty is the Command Broker; the shift cannot be delegated to an unlicensed person for the purpose of Gen AI tool governance.

7.1.3 MA-1 as AI Governance Failure Mode — Series First

The third CntlCo AI governance interaction point is a series of firsts: the MA-1 governing DBA-MA scenario (state estimator corruption via false data injection) is structurally identical to an adversarial ML attack on the state estimator's inference inputs.

The state estimator's ML components — bad data detection algorithms, topology processors, observability analysis tools — make inferences from measurement inputs. An adversary who injects false measurements at the right points in the network is, in effect, conducting an adversarial input attack on the ML system's inference. The attack does not target the ML system's weights or training data; it targets the inference inputs in real time. This is the same attack class studied in the adversarial ML literature — the adversary crafts an input that causes the ML system to produce a confident but wrong output.

This convergence has two consequences for the Dual-DB architecture. First, the AI Governance Design Basis for the CntlCo must include adversarial input robustness requirements for state estimator ML components that go beyond the random measurement error tolerance addressed in standard EMS bad data detection. The bad data detection performance specification must distinguish between random error tolerance (the standard) and adversarial injection tolerance (the AI governance requirement). Second, the MA-1 DBA-MA scenario and the AI governance failure mode analysis share the same underlying attack surface — the measurement input environment for the state estimator's ML components. Remediation measures that improve adversarial input robustness simultaneously improve both DBA-MA resistance and AI governance compliance. This is the only DBA-ES instrument where a single remediation measure satisfies requirements from both design bases simultaneously.

7.2 Bright Line Summary for CntlCo

CntlCo AI Application	Bright Line Position	Command Broker Requirement
State estimator ML-augmented bad data detection.	Below Bright Line — deterministic, formally testable	No Command Broker required for ML advisory output. Command Broker required if ML output triggers automatic redispatch without operator confirmation.
AGC optimization ML components	Below Bright Line — deterministic, formally testable	No Command Broker required. Bright Line condition: ML output limited to setpoint recommendations with operator confirmation; automatic execution without confirmation crosses Bright Line.
Gen AI situational awareness synthesis	Above Bright Line — non-deterministic	A licensed system operator on duty is mandatory for the Command Broker—delegation to a non-licensed person is not permitted.
Gen AI emergency procedure guidance	Above Bright Line — non-deterministic	A licensed system operator on duty is mandatory for the Command Broker. The operator must review the Gen AI

		emergency procedure output before execution.
Gen AI predictive dispatch recommendations	Above Bright Line — non-deterministic	A licensed system operator on duty is mandatory for the Command Broker. The operator must confirm the dispatch action independently of the Gen AI recommendation.

Section 8 — Acceptance Criteria (Updated for v0.3)

The acceptance criteria from v0.1 §8 are retained in full. The following criteria are added for the DBA consequence envelope and MA extension.

Criterion	Requirement	Derivation
SABD Zone 1 transition	State estimator recovery initiated within T_1 of confirmed non-convergence. Manual operations mode is fully activated within T_1.	SABD metric definition. T_1 calibrated to the NERC EOP-011-2 notification timeline at site-specific implementation.
SABD Zone 2 transition	RC mutual aid is established, and an emergency load shedding authority is confirmed within T_2 of SABD onset if Zone 1 recovery is not achieved.	SABD Zone 2 definition. T_2 calibrated to the cascade initiation timeline for the governing contingency set.
MA-1 bad data detection	State estimator bad data detection capability must be validated against adversarial injection scenarios — not only random measurement error. Adversarial injection tolerance specification required as part of AI Governance Design Basis.	§7.1.3 AI governance convergence on MA-1 attack surface.
MA-2 SABD persistence limit	Design basis maximum duration for MA-2 SABD Zone 3 conditions before mutual aid CntlCo support is invoked. Must be less than the estimated time-to-first-contingency under the governing threat scenario.	MA-2 secondary bounding scenario; §6.4.
IC-3 Tier 3 notification	Federal Coordination Threshold notification issued within [site-specific] minutes of confirmed IC-3 compound initiating condition.	DBA-EN-SF §3.4 Cross-Sector Consequence Elevation Rule trigger.
Gen AI Command Broker	No Gen AI tool output may be acted upon in real-time operations without confirmation by the licensed system operator on duty. Non-licensed personnel may not serve as	§7.1.2; DBA-ES §3.4.

	Command Broker for CntlCo Gen AI tools.	
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Section 9 — Open Items

The following items from v0.1 are closed by this instrument:

Item	v0.1 Description	Resolution
DI-1	DBA Consequence Envelope	CLOSED. Full DBA envelope established in §§ 3–4 of this instrument. SABD metric defined; IC-1 through IC-3 consequence taxonomy complete; Tier 1/2/3 designations applied.
DI-2	MA Extension	CLOSED. MA extension established in § 6. MA-1 governing scenario, MA-2 secondary scenario, four MA characteristics applied to unified envelope.
DI-3	RC authority chain communications design basis	CLOSED. RC authority chain communications independence requirement established in § 4.1: minimum one backup path physically and logically independent of primary SCADA/EMS communications infrastructure.
DI-5	SL-1 adversarial bad data injection design basis	CLOSED. Addressed in §§ 6.3 and 7.1.3. Adversarial injection design basis distinguished from random error tolerance; AI governance convergence on MA-1 attack surface established as a series of firsts.
DI-6	Dual-DB AI governance interaction points	CLOSED. Three interaction points established in § 7: state estimator ML augmentation, Gen AI operator decision support under Command Broker, MA-1/AI governance attack surface convergence.

The following items remain open:

Item	Description	Resolution Criterion
DI-4	Manual operations mode minimum capability specification. The minimum design basis duration for manual operations and the minimum functional capabilities that must be maintained in manual mode require site-specific calibration.	Established on the site-specific design basis implementation. Cannot be set at the series level.
DI-7	SABD T ₁ and T ₂ calibration guidance. Series-level guidance on calibrating T ₁ against the EOP-011-2 notification timeline and T ₂ against the cascade	CLOSED. Quantitative T ₁ and T ₂ series-level bounds established in DBA-ES-MA-QB v0.1 §2: T ₁ ≤ 30 minutes; T ₂ ≤ 4 hours. BA-class-differentiated recommended values in QB

	initiation timeline is provided at § 3.4; quantitative series-level bounds are not established.	v0.1 Table 2.1. §3.4 of this instrument has been updated accordingly.
DI-8	Sequential MTS attack CntlCo consequence. UC-2 OI-4 (sequential MTS attack) has CntlCo-specific consequences — the operator cognitive load of responding to a sequential attack that appears to be separate events before cascade onset is higher than for a simultaneous MTS attack. This scenario is not fully developed here.	Addressed in DBA-EN-UC-2 v0.2 addendum (UC-2 OI-4) and reflected in DBA-ES-CntlCo at next revision.
UC-2 OI-2	CntlCo coordination protocol for multi-node events. The coordination protocol between adjacent BAs and the RC during a multi-node attack scenario requires development beyond what is established in this instrument.	Deferred to DBA-ES-CntlCo v0.3 following UC-2 OI-2 development.

Section 10 — References

Regulatory and Standards References

North American Electric Reliability Corporation. BAL-001-2: Real Power Balancing Control Performance. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. BAL-002-3: Disturbance Control Standard. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. EOP-011-2: Emergency Operations. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. IRO-001-5: Reliability Coordination — Responsibilities and Authorities. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. CIP-008-6: Incident Reporting and Response Planning. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. CIP-014-3: Physical Security. Atlanta: NERC. Current version.

Government and Policy References

U.S. Department of Energy and North American Electric Reliability Corporation. "Final Report on the August 14, 2003 Blackout in the United States and Canada: Causes and Recommendations." Washington, D.C.: DOE, April 2004.

Federal Energy Regulatory Commission. Order No. 693: Mandatory Reliability Standards for the Bulk-Power System. Washington, D.C.: FERC, 2007.

Presidential Policy Directive 21 (PPD-21): Critical Infrastructure Security and Resilience. Washington, D.C.: The White House, February 2013.

Technical References

Kundur, Prabha. Power System Stability and Control. New York: McGraw-Hill, 1994.

Wood, Allen J., Bruce F. Wollenberg, and Gerald B. Sheble. Power Generation, Operation, and Control. 3rd ed. Hoboken: John Wiley & Sons, 2013.

Abur, Ali and Antonio Gomez Exposito. Power Systems State Estimation: Theory and Implementation. New York: Marcel Dekker, 2004.

Huang, Yilin, et al. "State Estimation Under False Data Injection Attacks: Theory and Defense." IEEE Transactions on Smart Grid. Vol. 10, No. 3, May 2019.

Series Parent and Cross-Reference Documents

Roxey, Tim. DBA-EN-SF: Energy Sector Design Basis Framework. Version 0.8 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES: Electric Sector Design Basis Framework. Version 0.8 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-CntlCo: Electric Sector Control-Company Functional Class. Version 0.1 — DRAFT. Eclectic Technologies, June 2026. (Predecessor document — DB layer.)

Roxey, Tim. DBA-EN-UC-2: Coordinated Multi-Node Attack on an Interconnection. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-DI-FC5: Distribution Substation with Edge Generation. Version 0.3 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-MA-SF: Electric Sector Military Action Sector Framework. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-MA-QB: Electric Sector Military Action Quantitative Design Basis. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial working draft. DB layer only. Three-layer design basis structure (PL, HL, SL) established. DBA consequence envelope and MA extension deferred to v0.2.
0.2	June 2026	T. Roxey	Full three-layer instrument. DBA consequence envelope added: three initiating conditions (IC-1 through IC-3), Situational Awareness Blackout Duration (SABD) metric established as CntlCo governing metric, Tier 1/2/3 consequence taxonomy applied. MA extension added: MA-1 (state estimator corruption) as governing scenario, MA-2 (SCADA availability attack) as secondary bounding event. Dual-DB AI governance interaction points detailed: state estimator ML augmentation, Gen AI operator support under Command Broker, and MA-1/AI governance attack surface convergence established as a series first. Open items DI-1, DI-2, DI-5, DI-6 from v0.1 closed.
0.2 (housekeeping)	June 2026	T. Roxey	Housekeeping correction (N-7). Series grandparent reference updated from DBA-EN-SF v0.6 to v0.7 in three locations: cover lineage line, §1.x body text (Series Grandparent entry), and Section 10 references list. DBA-EN-SF v0.7 adds SDD and SABD as fifth and sixth series-standard consequence metrics and updates the series inventory table—no substantive analytical changes.
0.3	June 2026	T. Roxey	N-15/N-16 batch pass (CntlCo anchor instrument). Four changes: (1) Series grandparent reference updated from DBA-EN-SF v0.7 to v0.8 throughout (cover, §1.3, §10 references). Closes N-7. (2) §3.4 T ₁ /T ₂ calibration paragraph updated to cite DBA-ES-MA-QB v0.1 §2 quantitative bounds: T ₁ ≤ 30 minutes, T ₂ ≤ 4 hours series-level upper bounds. Closes DI-7. (3) §6.1.1 MA Applicability Assessment block added: four-filter structured assessment per DBA-ES-MA-SF v0.2 §1.4 and QB v0.1 §5; CntlCo satisfies all four filters by definition; all-facilities conclusion stated. (4) §5.4 MA Communications Design Basis added: CntlCo-specific instantiation of CB-1 through CB-5 requirements per DBA-ES-MA-QB v0.1 §5; CB-1/SABD interaction chain established. FC5 reference updated to v0.3. DBA-ES-MA-SF v0.2 and DBA-ES-MA-QB v0.1 added to §10 references.
0.4	June 2026	T. Roxey	§7.1.1 advisory-only ML correction. Paragraph 1 rewritten to make advisory-only output the predicate of below-Bright-Line status, correcting a logical sequencing error in v0.3 where the below-Bright-Line assertion preceded its own governing condition. The below-Bright-Line determination is now stated as conditional (“Where ML output is limited to advisory flagging and state estimation inputs”) rather than as an inherent

			property of ML determinism. Paragraph 2 (the Bright Line condition statement) is unchanged. No other sections affected.
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